Unit 5 Using Files

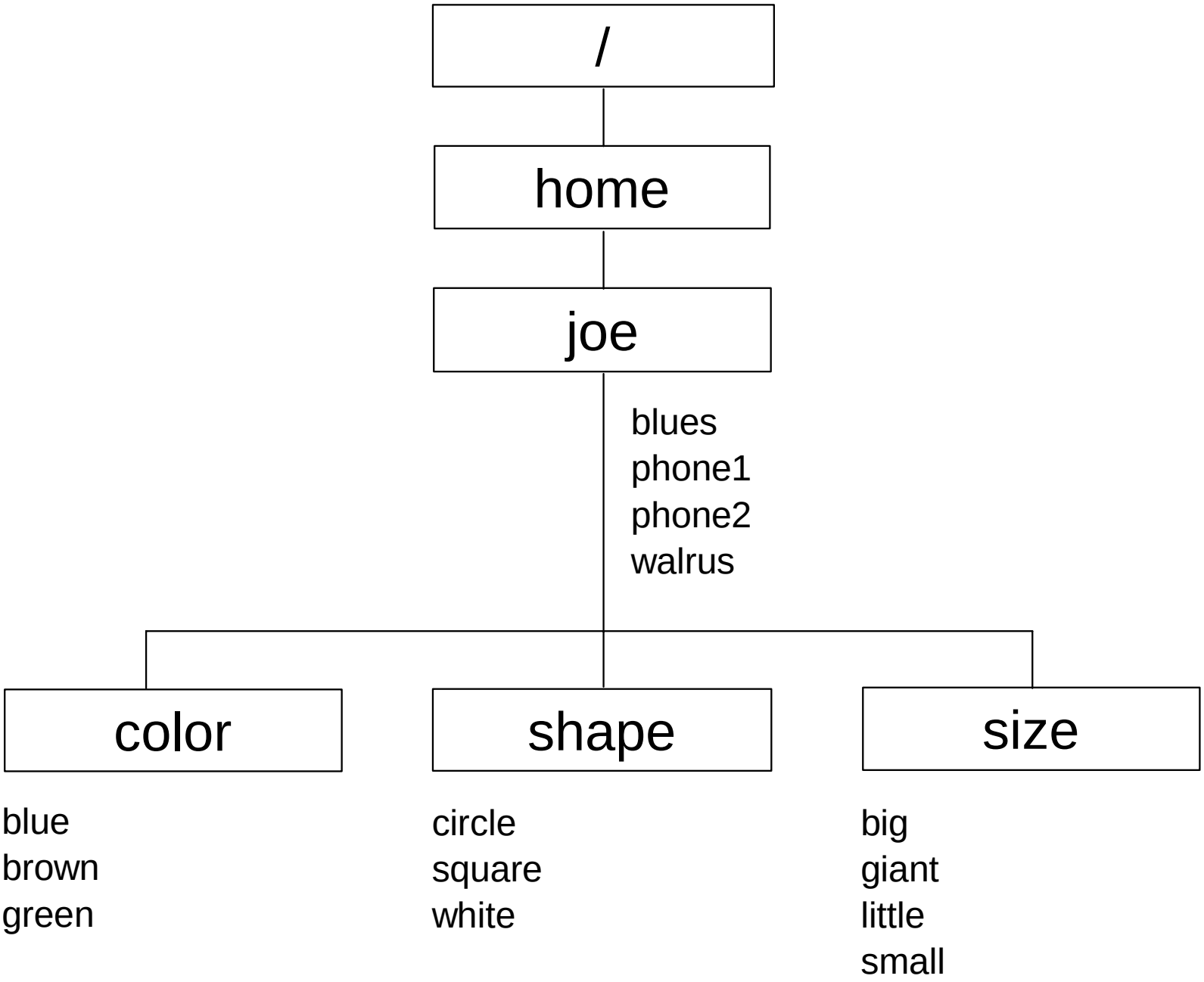
Course code LX13
Linux Basics

Objectives

After completing this unit, student should be able to:

- Use the `cp` command to **copy** files
- Use the `mv` command to **move** files
- Use the `ln` command to allow a file to have more than one name
- Display the contents of a text file using `cat`, `more` and `less`
- Use the `rm` command to **remove** files
- Use the `lpr` command to **print** files

Example Directory Structure



Copying Files

- With the `cp` command:

```
$ cp  source  target
$ cp  sourcefile1 sourcefile2 ... targetdir
```

- To copy the file **/home/joe/color/green** to **/home/joe** and name it **programa**:

```
$ pwd
/home/joe
$ cp /home/joe/color/green  programa
```

- To copy **/home/joe/shape/white** and **/home/joe/blues** to the **/home/joe/size** directory:

```
$ cp shape/white blues size
```

Copying Files *(2)*

- To copy **white** in the **size** directory to the **color** directory and use the same name:

```
$ cd size
$ cp white ../color
```

Moving and Renaming Files

- With the `mv` command:

```
$ mv source target
```

- To rename a file

```
$ cd ../shape
$ pwd
/home/joe/shape

$ mv white oval
```

Moving and Renaming Files *(2)*

- To move a file to a different directory and rename it:

```
$ cd /home/joe/size
$ pwd
/home/joe/size


$ mv white ../color/pale
```

- To move a file to a different directory, keeping the same name:

```
$ pwd
/home/joe/size


$ mv ../programa .
```

Linking Files

- With the `ln` command:

```
$ ln source_file target_file
```

- To refer to **/home/joe/phone1** as **phone_list** from the **/home/joe/shape** directory:

```
$ cd /home/joe
$ pwd
/home/joe

$ ln phone1 /home/joe/shape/phone_list
```

Listing File Contents

- With the concatenate command:
  `$ cat file1 file2 ...`

```
$ cat walrus
"The time has come", the walrus said,
  "To talk of many things:
Of shoes - and ships - and sealing wax -
  Of cabbage - and kings -
And why the sea is boiling hot -
  And whether pigs have wings."

From The Walrus And The Carpenter
by Lewis Carroll (1871)
```

Displaying Files

- With the `more` or `less` commands:

```
$ less walrus

"The time has come", the walrus said,
  "To talk of many things:

Of shoes - and ships - and sealing wax -
  Of cabbage - and kings -

And why the sea is boiling hot -
  And whether pigs have wings."

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From The Walrus And The Carpenter

by Lewis Carroll (1871)

(END)<q>
```

Displaying Binary Files

- With the `od` command:

```
$ od /usr/bin/passwd
0000000 042577 043114 000401 000001 000000 000000 000000 000000
0000020 000002 000003 000001 000000 107300 004004 000064 000000
0000040 051430 000000 000000 000000 000064 000040 000006 000050
.
$ od -c /usr/bin/passwd
0000000 177   E   L   F 001 001 001  \0  \0  \0  \0  \0  \0  \0  \0
0000020 002  \0 003  \0 001  \0  \0  \0   À 216 004  \b   4  \0  \0  \0
0000040 030   S  \0  \0  \0  \0  \0  \0   4  \0      \0 006  \0   (  \0
.
$ od -h /usr/bin/passwd
0000000 457f 464c 0101 0001 0000 0000 0000 0000
0000020 0002 0003 0001 0000 8ec0 0804 0034 0000
0000040 5318 0000 0000 0000 0034 0020 0006 0028
.
$
```

- With the `strings` command:

```
$ strings /usr/bin/passwd
/lib/ld-linux.so.2
__gmon_start__
.
```

Removing Files

- With the `rm` command:

```
$ rm file1 file2 ...
```

- To remove the **walrus** file in your current directory

```
$ ls walrus
walrus
$ rm walrus
rm: remove `walrus'? y
$ ls walrus
ls: walrus: No such file or directory
```

- Remove the walrus file non-interactively:

```
$ rm -f walrus
```

Printing Files

- To queue files to the printer `lpr`
- To display the status of print queues `lpq`
- To cancel a print job `lprm`

```
$ lpr /home/shape/square

$ lpq
Rank    Owner   Job Files     Total Size
active  joe     14 square     1066 bytes

$ lprm 14
```

Splitting Files

- If a file is too large (for instance to fit on a floppy) you can split it in a number of smaller files with the **split** command

```
$ ls -l
-rw-r--r--   1 root     root     10485760 Feb 21 13:31 largefile

$ split -b 1024k largefile largefile.

$ ls -l
-rw-r--r--   1 root     root     10485760 Feb 21 13:31 largefile
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.aa
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ab
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ac
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ad
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ae
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.af
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ag
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ah
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.ai
-rw-r--r--   1 root     root      1048576 Feb 21 13:33 largefile.aj
```

Checkpoint

1. Match the commands with the descriptions

A. cp	__	Remove files
B. ln	__	Copy files
C. mv	__	Rename files
D. less	__	Give a file a second name
E. rm	__	Print a file
F. lpr	__	Display a file one screen at a time

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Unit Summary

- The `cp` command can be used to copy files
- The `mv` command can be used to move and rename files
- The `ln` command can be used to create additional names for a file
- Display the contents of a file using `cat`, `more` or `less`
- The `rm` command can be used to delete files
- Use the `lpr` command to print files